iii. Hardware

Friday, 20 December 2019 3:05 PM

The hardware components on the floating table above the experiment bench are necessary to provide control of the experiment and to generate rf frequencies, laser frequencies and magnetic fields. Each component of the hardware as well as how these are interconnected is explained here.

Interlock



Above: The interlock on/off switch. Below: The interlock relay box.
The interlock switch allows any of the hardware connected to the interlock relay box to be switched off remotely from outside the lab. It contains a Galil logic controller which will trip to off in the event of a power outage, and must be manually switched back on. If this switch is on it relays a signal to the interlock relay box, on the northwest corner of the experiment table. This relay box takes the signal and relays it to each laser controller box to allow them to actuate laser radiation. In this lab the MogLabs, Toptica, and Keopsys lasers are connected to the interlock, as well as lasers on the other experiment table, not used for the BEC apparatus.
If the power is on to the relay box but the green light is not lit and lasers are not enabled, then check the interlock switch. If the lasers cannot be powered up, check they are connected to the interlock box from an output on their controller boxes marked interlock. If they are and there is still no power there may be an electrical fault within the interlock box and it should be taken to the Monash Instrumentation Facility for inspection. This is explained in detail in 'A. Hardware Troubleshooting'

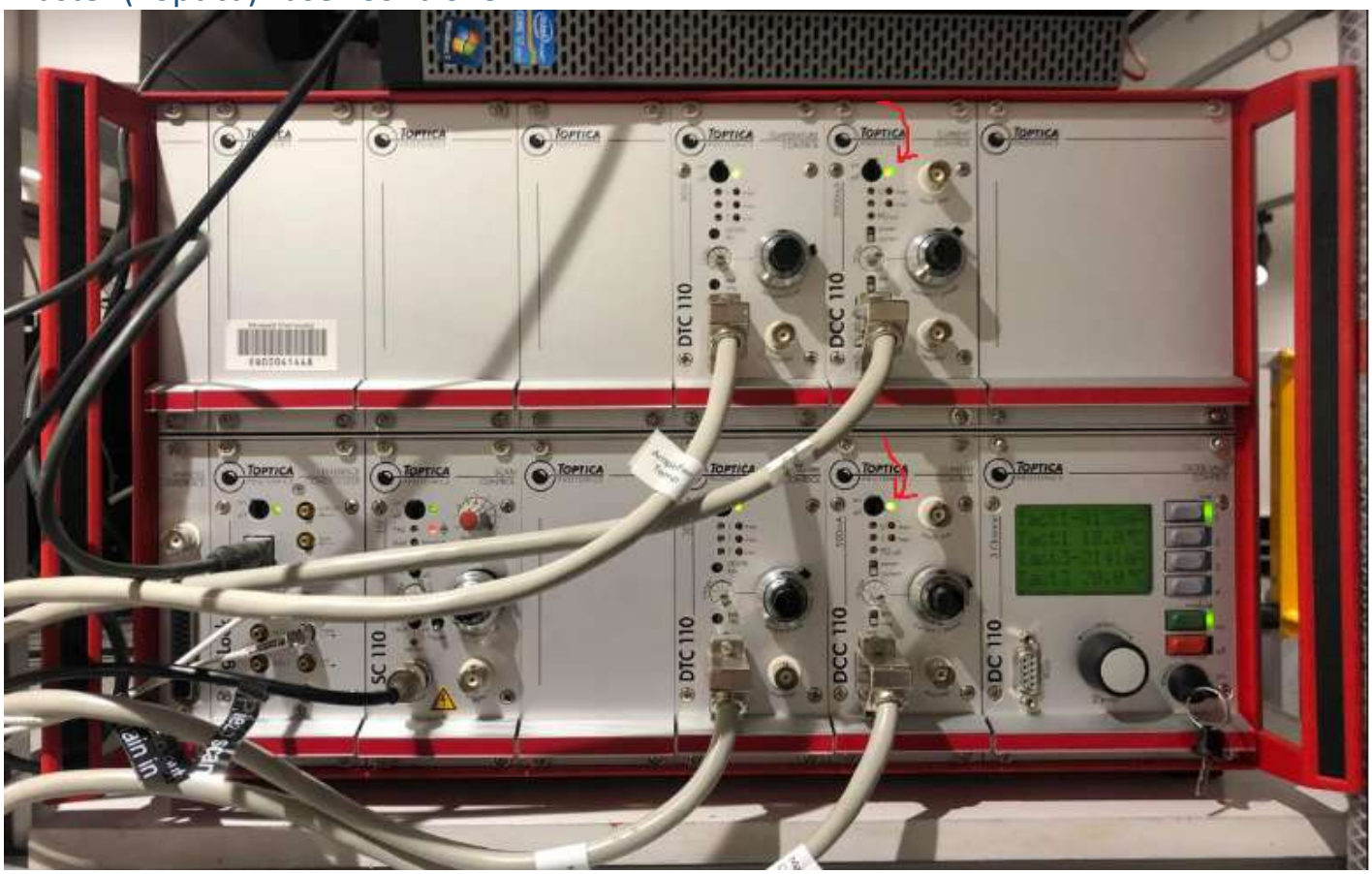
Repump (MogLABS) Laser Controller



The EDCL Diode laser controller, used to control and lock the frequency of the repump laser.

This controller is used to set the frequency of the repump laser to the required atomic transition frequency. It has a connected oscilloscope to view an absorption spectrum from the locking optical setup. It has input connections from the interlock, allowing power to be driven into the diode laser on the optics table. If the diode LED marked with a red arrow is red (and the two scan/lock switches are set to scan) then the interlock is preventing the laser from firing.

Master (Toptica) Laser Controller



The Toptica laser controller, used to control the tapered amplifier master laser and the side beam for locking the master laser.

This controller is used to set the frequency of the master laser to the required atomic transition frequency. The computer above the controller is used for locking the laser, equivalently to using the oscilloscope for the repump laser. The computer has no monitor and must be controlled remotely by VNC from the control PC.
This controller also has input connections from the interlock, allowing power to be driven into the tapered amplifier laser on the optics table.
Dials on the front can be used to change the current through the small locking laser (master current, on the lower half) or the larger current through the tapered amplifier (amplifier current, on the upper half). Details on locking both lasers are given in '7. Getting Started - Locking the Lasers'.

Dipole (Keopsys) Laser Controller



The Keopsys laser controller, used for controlling the power in the dipole laser beam

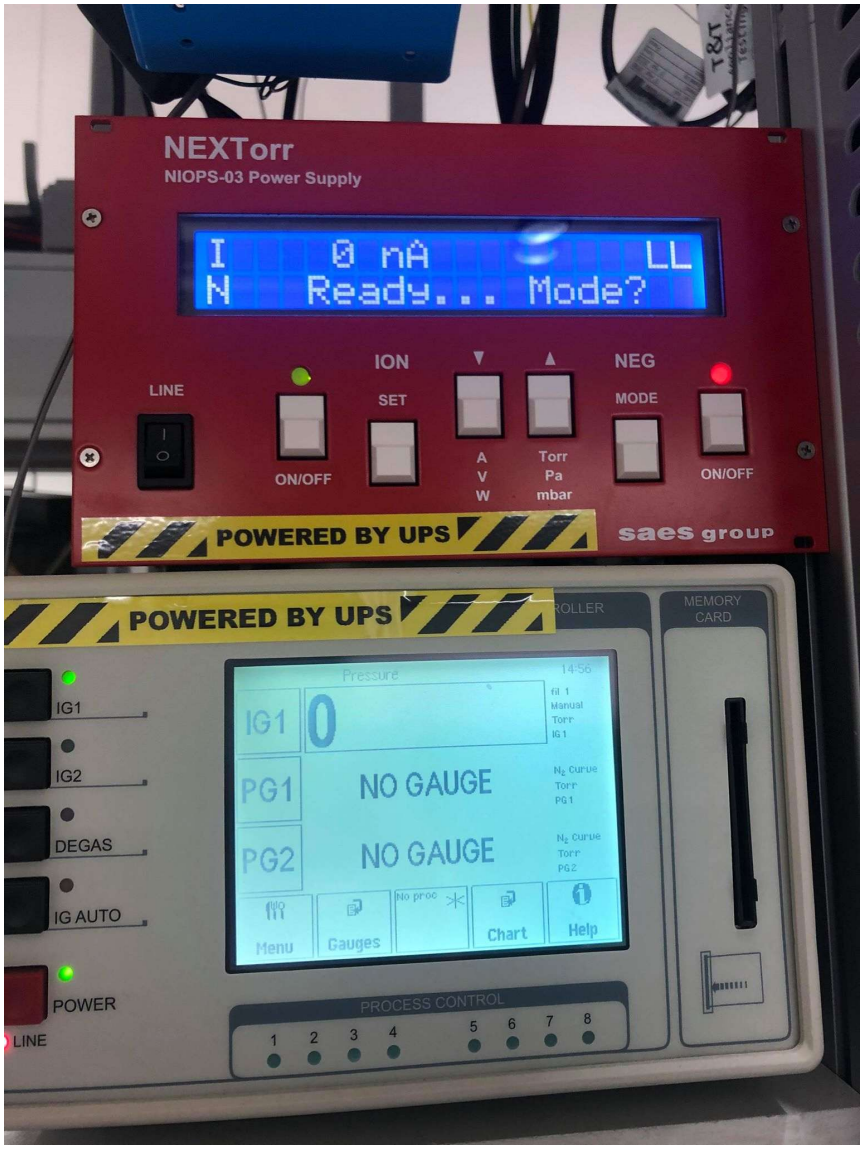
This controller generates the power for the 1064nm dipole laser, which is used to trap the atoms in a hybrid trap, where they are evaporated to make a BEC. Once turned on using the on switch and key, the 'pump' button is used to generate laser power. The front screen shows the output amplification of the laser, which can be tuned with the control knob to between 38dB (around 5W) and 43dB (around 20W).
This laser frequency is far off resonance so does not have a locking circuit. The only input into the box other than the power is the laser interlock, and otherwise the controller is completely self-contained.

UPS (Uninterruptible Power Supply)



The UPS is a device for providing backup power to certain essential hardware in the event of a power outage. Its located above the second experiment bench, to the northeast corner of the lab. It charges its batteries from the mains and can then be used for backup power for a couple of hours if required. An output gang cable from the UPS runs to the main experiment table. Plugged in are the master and repump laser, the ion and NEG pumps and a pressure monitor. It is critical to keep constant power to the pumps to prevent the pressure in the science chamber increasing and potentially taking weeks to pump down, if the pressure becomes high enough that it causes a leak in one of the seals.

Pump Power Supply and Pressure Gauge



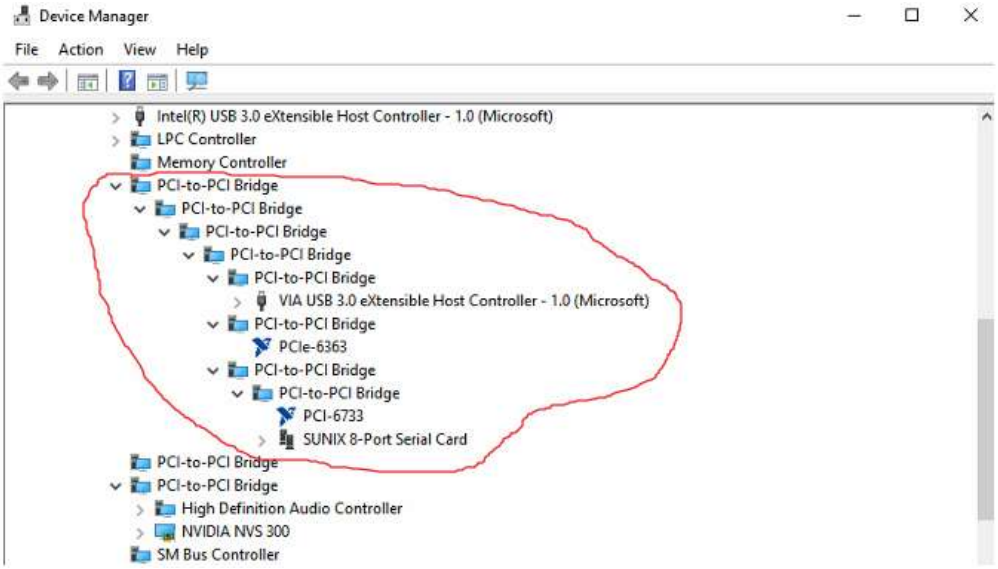
As considered above, these electronics are powered by the UPS and should read values of 0 nanoamps, and 0 torr, when the getters or UV lamps are not in use. The lower ion gauge is not a pump power supply, it is only used to monitor the pressure. It has a limit of 1e-12 torr, far more sensitive than the NEXTorr pressure sensor, so a reading of 0 indicates the pressure is below its minimum resolvable pressure. Equally, 0nA current measured by the Nextorr pump controller is effectively a measure of pressure, via the number of particles flowing into the pump. The NEG pump should show a red light as this pump runs in passive mode, with molecules adsorbing to its large surface area. The ION pump should show a green light as it actively pumps the molecules out, and is most needed to remove nobles gases in the chamber that will not adsorb to the NEG pump.

PCIE Expansions - Fibre range extender



All of the hardware placed on and above the experiment table can be remotely controlled via the control and imaging PCs behind the shielding screen. As it would be highly inconvenient to run long cables directly to the PCs from the table for each controlled component (e.g. camera, National Instruments card), a PCIE expansion is used. The computer case shown above contains expansion cards for the Control PC. An orange optical fibre runs from the control PC into this case, meaning it is a fibre range extender. In this case there are then USB ports into which ThorLabs cameras can be plugged, for imaging the 2D MOT, for example. It also has outputs to control the National Instruments cards and PulseBlaster, which in turn are used to control the hardware.

If the BLACS software on the Control PC cannot actuate the hardware, it may be because the control PC does not 'see' this extender. To ensure the control PC can see it, the extender needs to be turned on BEFORE the control PC. The control PC is then fully turned off and unplugged. Then, the range extender needs to be turned off and on again in 60 seconds (unplug the mains cable too) and then the control PC turned back on once the extender is on. This is known as power cycling. Repeat this process until the computer can detect that the range extender is connected. This connection is visible in Device manager -> View -> Devices by connection as a nested series of PCI-to-PCI bridges with the NI cards at the end:



Equally, the computer case below was the fibre extender for the imaging PC. It was used to indirectly plug the side and bottom cameras into the imaging PC, but developed unresolvable connectivity issues. It worked in the same way as the control PC extender but for the side and bottom cameras. It has been replaced by running long wires to the imaging PC; an ethernet cable for the side camera and a power-boosted USB for the bottom camera.



As the USB would lose power over a long, roughly 5-10m cable, these ICYBOXes can be used to boost the signal greatly and allow hardware to be connected across the lab without power loss. One is also used to connect a 2D MOT monitoring camera to the control PC PCIE expansion box.



PulseBlaster



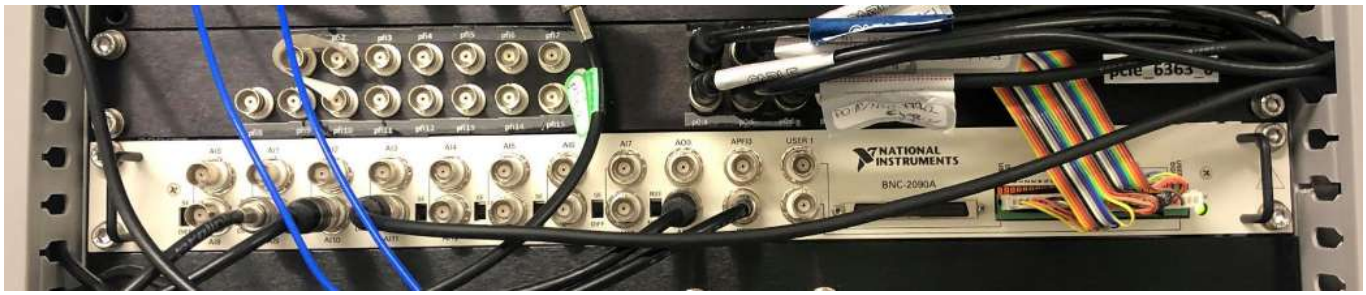
The PulseBlaster module acts a pseudoclock to control the precise timing of the experiment. A pseudoclock is a clock which ticks only when an output needs to change state, otherwise known as a variable frequency clock. It updates at the fastest rate needed for the quickest instructions needed during the run (usually the ramp during the magnetic catch), and relays the timing instructions to certain devices. The National Instrument cards and NovaTechs which physically control the hardware are commercial devices, and do not have large enough on-board memory to contain the instructions themselves. By offloading the timing requirements to the PulseBlaster, those memory limitations are removed, as the devices can receive instructions only when their output should change.
There are 3 timing outputs from the PulseBlaster, for the NI cards, the NovaTech_0 and NovaTech_1 for port 0, 1 and 2 respectively.
The Pulseblaster module also generates some of the RF frequencies for the AOMs, specifically the two which control the dipole laser. This is necessary because the AOMs are still low power enough for the PulseBlaster, but the PulseBlaster updates much faster than the supernova, so the AOM output can be changed much more quickly.

[AST] Changed the Pulseblaster set-up by adding a second PulseBlaster - Details Day 25: Added a second PulseBlaster. This was done because of the need of kicking and the Raman lasers require sum 100ms resolution (100ms is the step size for NovaTech). Now

- Pulseblaster_1 is slave is mounted next to pulseblaster_0
- Pulseblaster_0 --> flag 3 to trigger the slave pulseblaster_1
- Flag 0 of slave pulseblaster_1 --> trigger the side camera

National Instruments (NI) Acquisition Cards







These acquisition cards are designed to control the hardware of the experiment, by communicating with drivers that actuate the hardware devices to work, such as to open and close the shutters, and generate rf signals. These cards are plugged directly into the Control PC range extender. They receive control signals from BLACS, a component of the experiment software on the Control PC, and

then use coaxial outputs to connect to the devices they provide signals to, either analogue outputs, or digital TTL signals. Equally, some cards have coaxial inputs to receive data from hardware and transmit this data back to the Control PC.
As mentioned above, the precise timing of executing instructions on the hardware is controlled by the PulseBlaster.

There are two instances of the first card, the NI pci_6363 on the apparatus. These are multifunction cards which have both digital and analogue outputs and inputs. The 6363 card on the south side of the table has digital outputs controlling:

- Certain AOMs, on/off
- The LEDs used to desorb rubidium
- The laser shutters
- The Sorensen power supply to the Science coils
- The bottom camera trigger

The 6363 card on the north side also has digital outputs, controlling:

- Certain AOMs, on/off
- The servo to switch between imaging and MOT creation
- The bottom imaging mirror motor on/off

It also has an analogue output to control the current through the rubidium getter.

The majority of analogue inputs are also on this north side card. These are to monitor the current through the quadrupole coils and the strength of the bias field in the Z axis, and also MOT fluorescence (using a photodiode) and IGC pressure.

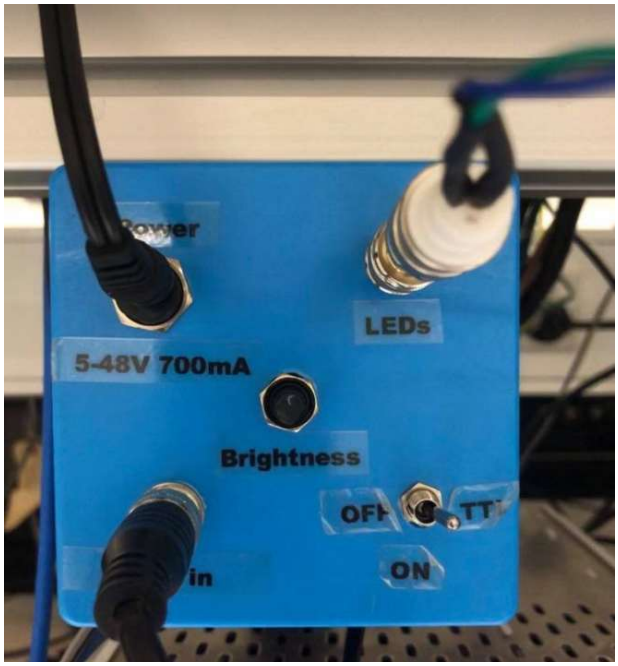
The pci_6733 is an analogue output card, allowing the strength of the magnetic fields in both the Source and Science MOTs to be controlled. Signals relayed from the control PC, along the fibre range extender and into the PCI_6733 card cause the card to actuate the Magneato drivers responsible for each coil.

Shutter Drivers



This driver case contains 8 Uniblitz ED12DSS bistable shutter drivers. These electronics allow up to 8 Uniblitz TS6B laser shutters to be opened and closed using TTL signals. On the front are TTL input ports connected to the pci_6363 card, and a DPDT switch allowing the shutters to be manually on, manually off or TTL controlled remotely. On the rear, coaxial cables connect the drivers to the shutters placed on the experiment table. In TTL mode, the shutters can be actuated using the BLACS user interface on the Control PC.

Source MOT LED Controller



This simple box allows control of the UV LEDs in the source chamber. These LEDs were added to prevent the atom number in the MOT dropping throughout the day as rubidium atoms adsorb onto the metal walls of the chamber, and the getter is fired only once in the morning. The LEDs are switched on while the 2D MOT loads into the 3D MOT, giving higher atom numbers in the 3D MOT. There is an input power line, a DPDT switch for on/off/TTL control, a power output to the LEDs and a TTL trigger line. The trigger comes from the NI 6363 card, so can be remotely controlled from BLACS.

SuperNova RF generators



These cases each contain a Novatech DDS9M signal generator, capable of creating the rf signals needed to run AOMs and perform forced evaporation with an rf 'knife'. There are four outputs of each device, though two are static and cannot be changed during a run of the experiment. The modules are called SuperNovas as they also contain necessary electronics such as thermal switches and relays, as well as the DDS9M signal generators.
Similarly to the shutter drivers, they have an input for TTL control from the pci_6363 card, and a DPDT switch to turn the generators to manual on/off or TTL control.

The output signal is either through a 'test' line, for debugging to ensure the internal electronics are working correctly using an oscilloscope, or to the 2W amplifier line 'AMP' which connects directly to the AOMs. Ensure all AOMs in use are set to AMP to ensure they are receiving rf power at the correct frequency.

The SuperNova receives instructions of how to control each AOM from the control PC via a VGA cable input (white wire) from the control PC PCIe expander. These instructions, which set frequency and amplitude, are either executed immediately in manual mode, or are queued up at the start of a shot, ready to be executed.
The SuperNova also has a clocking signal input from the PulseBlaster. This tells the SuperNova the time increments to step through instructions for hardware timed channels which were queued up from the control PC.

In the file 'ConnectionTable.py', accessible through BLACS, the outputs of each SuperNova are listed. This information may also be written on the wires plugged into the 'AMP' out coaxial connections.

RF Power Amplifier



This device is currently disconnected, though may be used when needing to provide a stronger rf signal, such as for some more exotic modes of trapping the BEC. It can be connected in series between a SuperNova output and the rf device being powered.

Getter Power Supply



This power supply provides current through the rubidium/potassium connector to boil atoms from the getter surface and fill the source chamber with atoms for the experiment. The output wires connect to the getter driver, housed in the same box as the source coil Magneato drivers.

Source Coil Magneato and Power Supply



The black box contains four independent Magneato drivers to individually address the North, South, Top and Bottom coils that form the 2D magneto-optic trap. Each coil driver is individually optimised for the resistance and inductance of the connected coil to ensure optimal performance, which is achieved by changing the value of some key resistors and capacitors on their circuit boards. Current is provided by the power supply, above, showing 33A split across four coils. There is a input control coaxial cable from the NI_6733 card to each driver, as these coils require analogue control. The drivers simply measure the current through the coils, compare this to the control voltage sent by the lab user, and adjust accordingly to ensure the current follows the control value.

If there is ever an issue with the coils, such as no current flows through the coils or the IGBT is suspected to be fried, please refer to the section 'B. Coils Troubleshooting'.

The getter driver, with outputs to the right, is controlled by the NI_6363 card, and draws power from the getter power supply mentioned previously. The outputs can be physically connected to any of the 8 getters (2 in each of 4 arms) on the experiment table, to switch between rubidium and potassium, or in the event a getter becomes spent.

Bias fields Power supply

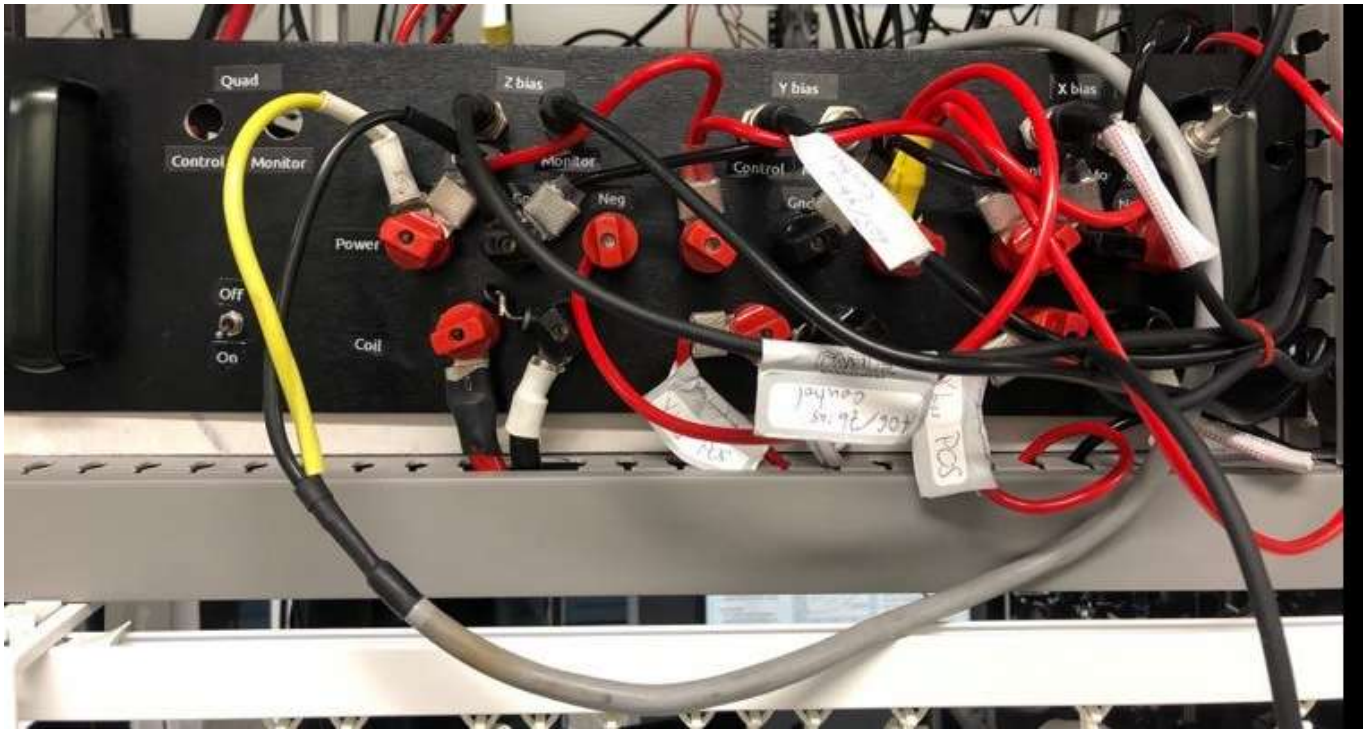


These power supplies provide current to the Science MOT bias fields, used to steer the position of the field zero of the powerful Quadrupole coil magnetic field. They can also be used to cancel out stray, background magnetic fields when performing Polarisation Gradient Cooling (PGC), when the Quadrupole Coils are switched off.

The top 'Manson' power supply is the negative power supply. It connects to the negative input ports on the Science coil magneato box. When the control current provided to a magneato driver from the control computer (via the ni_6733 card) is made negative for either the X or Y or Z bias coils, the current is drawn from this power supply and into the coils!
The bottom 'DCSS' power supply is the positive power supply. Likewise, it connects to the three positive input terminals of the science coil magneato and when any of the X,Y or Z coils receive a positive control voltage from the computer, the current is drawn from this supply to power the coils. In this configuration, the current can then flow both ways through the bias coils!

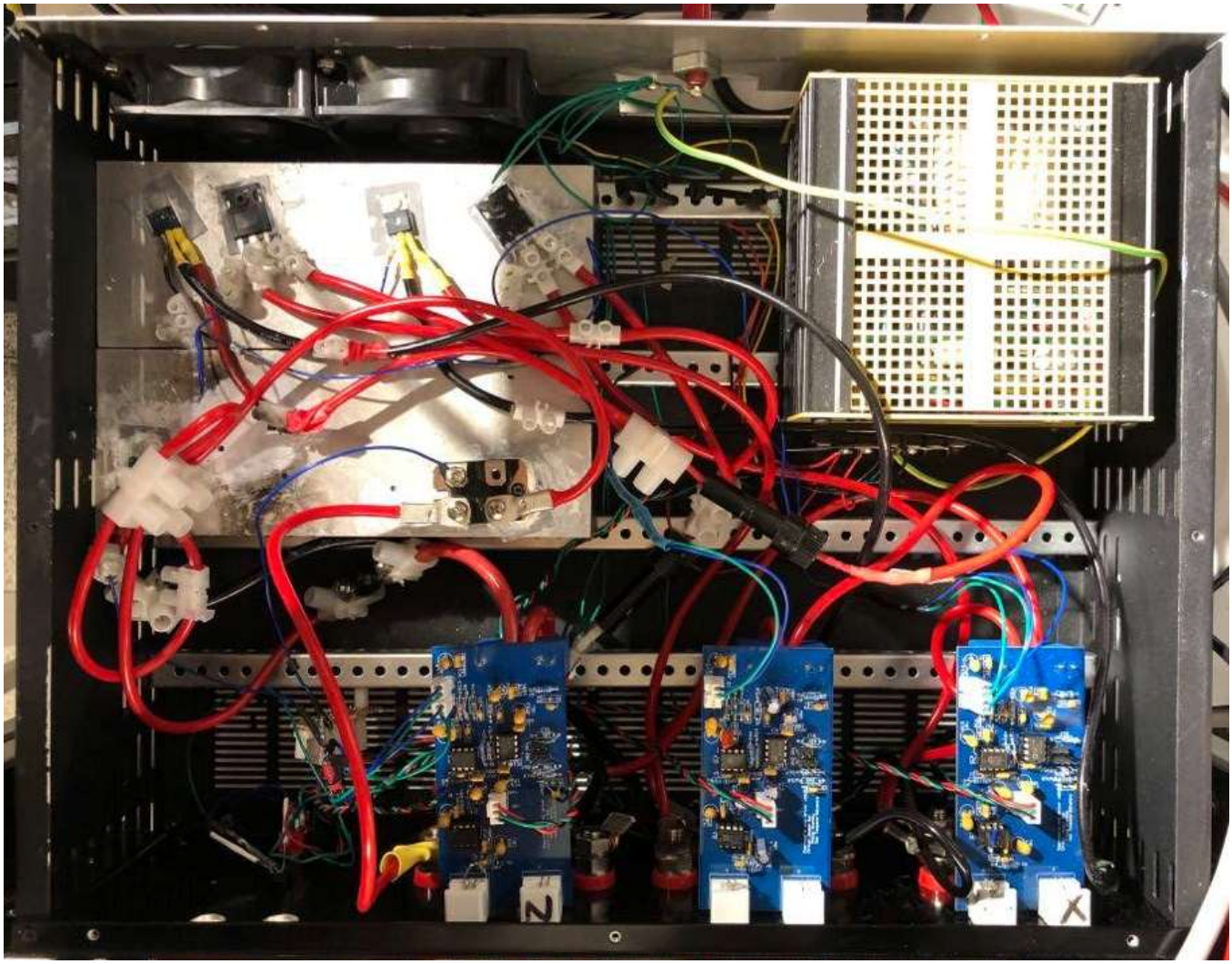
This is not the case for the source coils or the quad coils, as these have only a single power supply each, and current will always flow in one direction. This makes sense as the pairs of source coils can be individually addressed to change the field zero location, whereas the science bias coils are controlled together in pairs (two X coils, two Y coils etc.)

Bias Fields Magneato Driver



This box contains three individual magneato coil drivers, used to individually address the three pairs of bias coils used in the science MOT to translate the position of the quadrupole field zero. The ON/OFF switch allows +-15V from the box power supply to connect to the coil driver boards and power them. Therefore the coils cannot work without this switch, and it is good practice to switch it off at the end of the day to prevent coil overheating. As mentioned above, the DCSS power supply connects to the positive input terminals of each controller, and the Manson connects to the negative inputs. There are then output connections below where the coils are directly connected to the driver box.

At the top of the above picture, there is a row for input control signals from the ni_6733 card via BNC connector (marked 'Control') and a connector next to it for monitoring the current through the coils. This works as the drivers each have a Hall Probe attached which one of the coil wires passes through. The current through the coils is then sensed by the Hall Probe due to its magnetic field, and the probe generates a current 2000x smaller than the coils current. Using a well-chosen sense resistor, the voltage on the sense resistor can be between 0 - 7.5V depending on the coil current. The voltage across this resistor is then read out using the 'Monitor' connector, and the data collected by the Ni-6363 card.



Quad Coils Power Supply (Sorensen)



This large power supply is located outside the experiment room due to its loud cooling fan. The Sorensen has thick power cables, capable of delivering far above 100A of current to power the Quadrupole coils. These wires travel across the lab and feed into the quad coil Magneato box, at the rear, via a grey Anderson connector. It is also connected to the quad coil interlock box. If the interlock senses that cooling is not flowing through the coils, or that the temperature in the coils is too high, it prevents the Sorensen from generating current.

In standard MOT mode, the Sorensen should be in its low voltage mode at 5V, and roughly 7A of current flowing through the coils (equivalent roughly to a quad control voltage of 0.45A in BLACS).

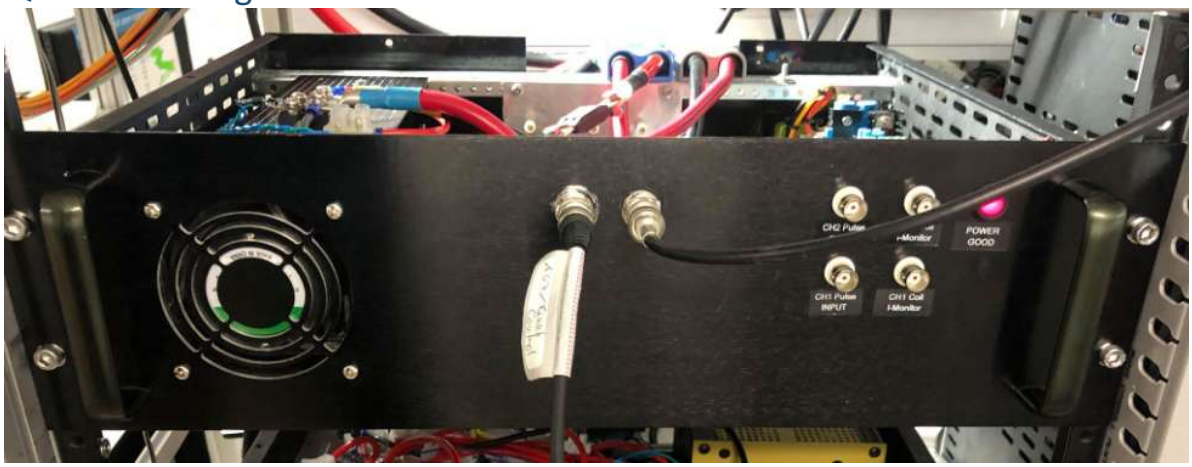
If the Sorensen shows no voltage or current (as above) then the interlock is disengaged (press the 'Quad Interlock on' button on the quad interlock box). If there is the 5V present but no current is flowing (and a non-zero control voltage is provided) then the circuit is open (coils are disconnected, IGBT circuit may be damaged etc.). This is covered in 'B.Coils Troubleshooting'.

Quad Coils Interlock



This case contains electronics to process information about the temperature and flow rate of water through the Quadrupole coils. This is necessary to prevent thermal runaway in the coils, which would be disastrous as they would have to be rewound and the surrounding optics removed to allow this. Similarly, a water leak could damage electronics in the lab. Inside is a Galil with 4 temperature inputs which show green lights when within acceptable temperature ranges. Probes attach to the quad coils where they meet the power wires from their Magneato, with both RS Pro flow meters and temperature sensors used. Pressing the green 'Quad Coil Interlock' button will both allow the Sorensen power supply to provide current through the coils, and switch on the pump to allow water to flow in the coils. This will turn the State LED green. The interlock should be switched off at the end of each day using the lower red button, to prevent water flowing when the lab is unsupervised.
The Z bias coil Interlock takes up the right-hand of the case, but is not currently used. It was designed in case stronger Z bias fields were needed for more exotic experiments. In the current configuration, the currents flowing are not high
enough (max 20A) to require temperature monitoring and water cooling.

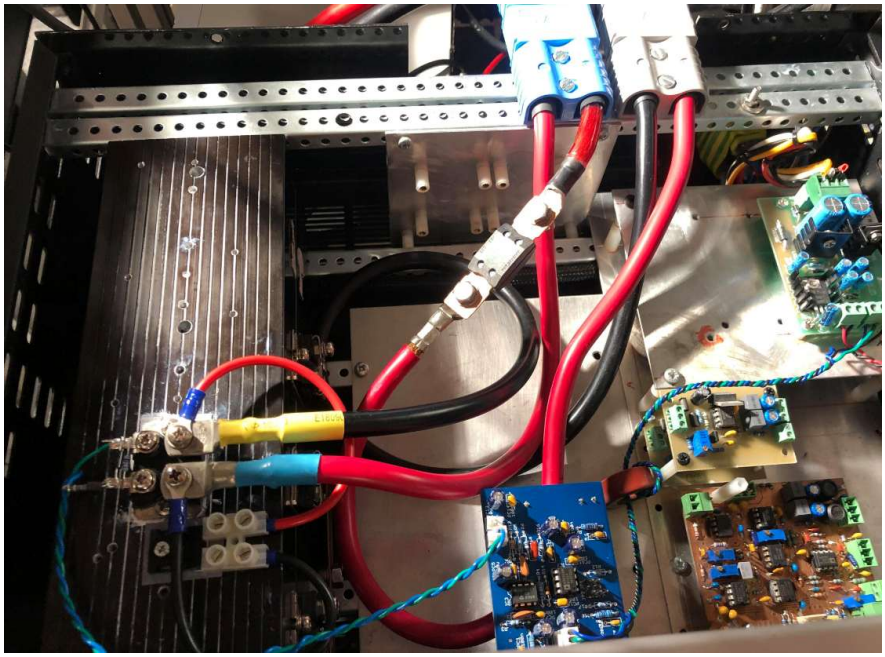
Quad Coils Magneato



This box contains the Magneato driver for the quad coils, along with an IGBT to physically change the current through the coils, and the driver power supply. A control voltage is fed into the front of the box via BNC cable from the NI 6733 card. This voltage passes into the Magneato driver, which has an attached hall probe to measure the current flowing through the coils. Internal electronics generate a corresponding voltage to this current (the sense voltage) and compare it to the control voltage. If the voltages are different, the circuit changes the current through the coils to match the control voltage.
It does this by changing the output voltage over its gate wire. The gate wire is attached to the gate of

the IGBT, basically acting as a variable resistor. Changing the gate voltage changes the amount of current that the IGBT will pass.
The sense voltage is also read out by BNC connector from the front of the box. The current through the coils can then be monitored during a shot using lyse (the **Coil_voltage.py** routine).

When switching the current fast, such as during the catch, the IGBT heats up appreciably via Ohmic heating, as power is dissipated in the IGBT in order to prevent too much current flowing in the quadrupole coils. Therefore, the IGBT is mounted on a fan forced heat sink. Even so, when running multiple shots, we found the IGBT to still heat to up to 100 degrees (measured via IR thermometer).



Internal layout of the quad coils Magneato. The Sorensen power supply is connected via grey Anderson connector, and the quad coils via blue connector. The Magneato driver receives +-15V power from the box power supply, and connects to the gate of the IGBT, to vary the current through the coils.

If any issues arise with the quad coils (as happened to us), read the section 'B. Coils Troubleshooting', for explanations and troubleshooting tips. As a quick test, place the clamp ammeter across a coil wire (outside the box) and measure the current. For a control voltage of 0.45V, the current should be around 7A. Run a shot and observe the current on the ammeter (or in lyse via the **Coil_voltage.py** routine). If the current is zero then there may be an issue (as long as the Sorensen is turned on and the interlock is engaged).